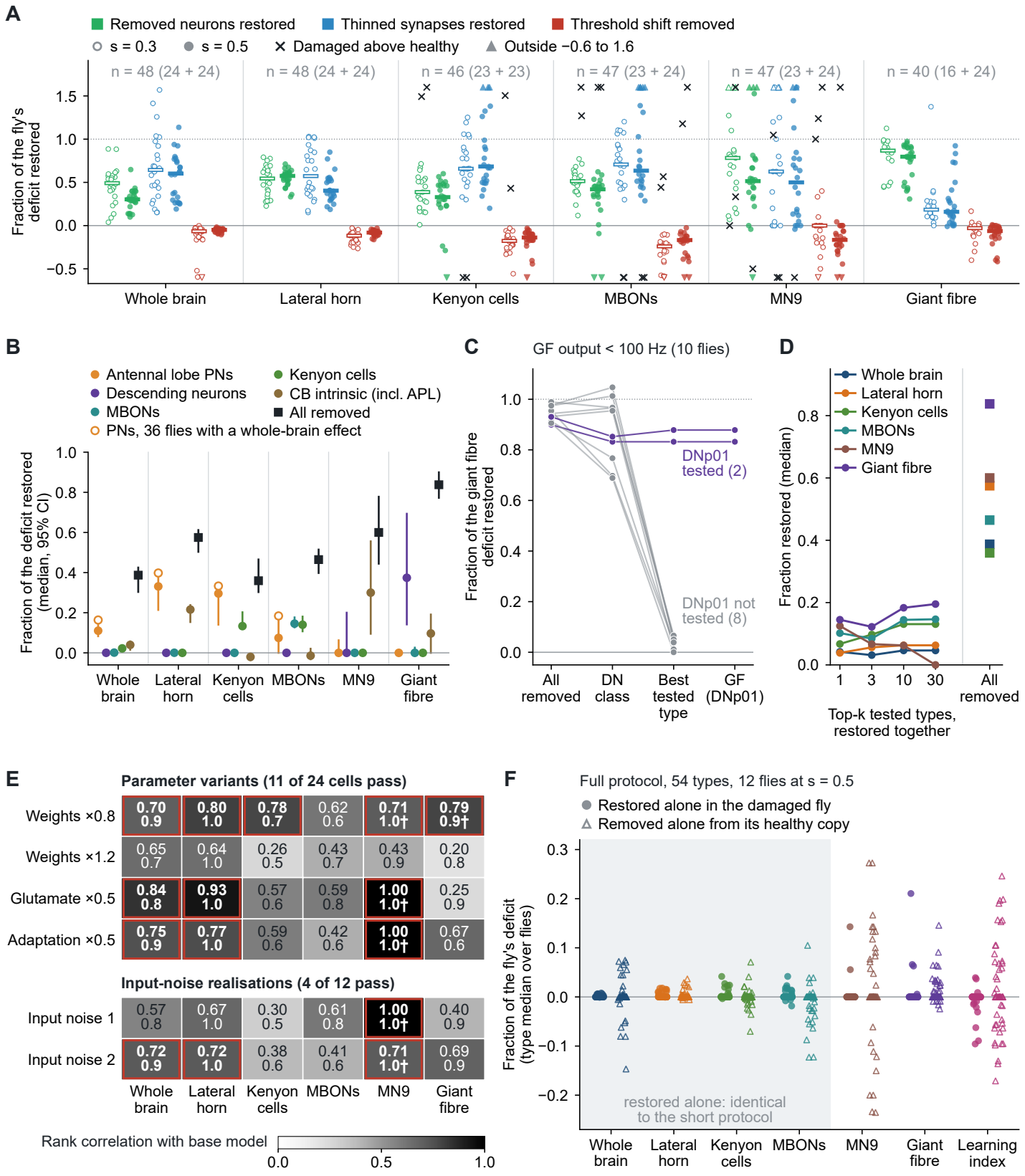

A
Removed neurons restored
Thinned synapses restored
Threshold shift removed
s = 0.3
s = 0.5
Damaged above healthy
Outside −0.6 to 1.6
n = 48 (24 + 24)
n = 48 (24 + 24)
n = 46 (23 + 23)
n = 47 (23 + 24)
n = 47 (23 + 24)
n = 40 (16 + 24)
Fraction of the fly's deficit restored
Whole brain
Lateral horn
Kenyon cells
MBONs
MN9
Giant fibre
B
Antennal lobe PNs
Descending neurons
MBONs
PNs, 36 flies with a whole-brain effect
Kenyon cells
CB intrinsic (incl. APL)
All removed
Fraction of the deficit restored (median, 95% CI)
C
GF output < 100 Hz (10 flies)
Fraction of the giant fibre deficit restored
DNp01 tested (2)
DNp01 not tested (8)
All removed
DN class
Best tested type
GF (DNp01)
D
Whole brain
Lateral horn
Kenyon cells
MBONs
MN9
Giant fibre
Fraction restored (median)
Top-k tested types, restored together
All removed
E
Parameter variants (11 of 24 cells pass)
Weights ×0.8
Weights ×1.2
Glutamate ×0.5
Adaptation ×0.5
Input-noise realisations (4 of 12 pass)
Input noise 1
Input noise 2
Rank correlation with base model
F
Full protocol, 54 types, 12 flies at s = 0.5
Restored alone in the damaged fly
Removed alone from its healthy copy
Fraction of the fly's deficit (type median over flies)
restored alone: identical to the short protocol
Learning index